



NAUVAL ZULFIKAR

Applied ML / NLP Engineer · Responsible AI candidate Bandung, Indonesia (relocatable to Saarbrücken, flexible start date) zulfikar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264 [LinkedIn](#) · [GitHub](#) · [Portfolio](#) **Visa status:** UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

MSc Business Analytics graduate from Aston University with a First Class degree, specialising in NLP and ML applications. My dissertation, supervised by Dr. Ammar Al-Bazi, focused on blockchain-integrated LLM analysis for supply-chain customer reviews, providing a strong foundation for research in LLM theory and interpretability. Extensive experience in applying ML in banking, government, and start-up environments, with a keen interest in transformer interpretability and computational cognition.

EDUCATION

Aston University, Birmingham, UK (*MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024*)

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan (*BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019*)

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- Distinctions: Business Data Analysis, Consumer Behaviour, Marketing Research

Indonesia AI (*NLP Engineer Training · EQF Level 5 · May – Nov 2023*)

- Contributed to the *2024 Indonesia Election Aspect-Based Sentiment Analyser* project
- Distinctions: Python Programming, Text Summarisation, Voice Recognition

Qarir Generator (Dubai, UAE) (*Data Science Global Accelerator · EQF Level 5 · Apr – Jun 2023*)

- Led sentiment-analysis development for the *Pfizer Public Sentiment Analysis* project
- Distinctions: Python, SQL, Pandas/NumPy

RESEARCH-RELEVANT PROJECTS

LLM-Generated Adaptive Shipper Decision Rules (2026)

- Developed an end-to-end RAG system using gpt-4o-mini augmented with retrieval over a curated behavioural-literature knowledge base. Focused on reproducibility and ablation.
- github.com/nauvalZulfikar/LLM-Shipper-Profiles

MSc Dissertation Pipeline (DeBERTa-v3 fine-tune) (2024)

- Fine-tuned DeBERTa-v3 on multi-tier supplier review data for multi-aspect classification. Evaluated emergent biases and integrated into a decision system.
- Graded 1:1.

Crowd-Shipping ABM (450-run behavioural parameter sweep) (2026)

- Conducted a controlled experiment over reward and supply-density grids, achieving acceptance rates from 45.34% to 79.80%.
- github.com/nauvalZulfikar/Crowd-Shipping-ABM-

VRPOD MILP Optimisation (2026)

- Implemented MILP optimisation for VRPOD, achieving a 38.5% cost reduction on a Birmingham-inspired instance.
- github.com/nauvalZulfikar/VRP-MILP

Mini Digital Twin (Integrated Passenger-Freight Transport) (2026)

- Developed a three-layer Digital Twin prototype with interactive dashboards and simulation capabilities.
- github.com/nauvalZulfikar/Mini-Digital-Twin

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung (*Data Scientist & Project Manager · Sep 2022 – Present*)

- Automated ETL pipelines and built self-service dashboards, achieving 45% faster project delivery and 20% data-efficiency gain.

Syncwell, Birmingham, UK (*Marketing Data Scientist · Jul – Sep 2024*)

- Implemented predictive ML and A/B testing, resulting in a 40% CTR increase and 22% follower growth.

PCOS Challenge, Birmingham, UK (*Strategic Data Scientist Specialist · Jun – Aug 2024*)

- Developed a customer attribution model and automated report extraction, reducing manual workload by 80%.

Bank Muamalat Indonesia, Jakarta (*Lead Data Scientist · Nov 2019 – Aug 2022*)

- Developed NLP models for credit analysis and risk scoring, achieving a 5.3% investment return.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Working paper on ML regressor evaluation for retail sales forecasting.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API + RAG, sentence-transformers, classification, regression, time-series forecasting, recommender systems, attribution modelling

Languages/Infrastructure: Python, SQL, R, PySpark, NumPy, Scipy, pandas, scikit-learn, FastAPI, Streamlit

Mathematical/Formal: MILP (Pyomo, scipy.optimize.milp, HiGHS), Agent-Based Modelling (mesa), Discrete-Event Simulation (simpy), Digital Twin architecture, Bayesian methods, causal inference

Governance: Model validation, model reliability, audit-grade documentation, model staleness monitoring, A/B testing, AWS, Docker, Git, ETL/orchestration, Power BI, Tableau

LANGUAGES

Indonesian (*Native*) · **English** (*Fluent, IELTS 7.0 overall: L7.5/R7.5/W6.5/S7.0*) · **Japanese** (*Intermediate*)

REFEREES

Dr. Ammar Al-Bazi

Senior Lecturer, Aston Business School, Aston University

Email: a.al-bazi@aston.ac.uk

Relationship: MSc dissertation supervisor (2023-2024)

Dr. Viktor Pekar

Lecturer, Aston Business & Social Sciences

Email: v.pekar@aston.ac.uk

Relationship: MSc Personal Tutor; taught Machine Learning for Business Analytics + Research Methods